

Project Title (≤ 20 words) *

Detecting conflict-induced deceptive behavior through representation geometry

Plain-language Summary (2-3 sentences) *

Use plain, concrete language aimed at a general technical audience.

We focus on conflict-induced deceptive behaviors: cases in which conflicting information creates pressure for the model to verbalize something different from what its internal evidence best supports. We study three categories of conflict: (i) conflicts between a model's prior belief and a user prompt, (ii) conflicts within a prompt or retrieved context, and (iii) conflicts between an agent's private evidence and public coordination in multi-agent systems. We will design white-box detection methods for representation geometry and compare them against strong black-box and human-behavior-derived baselines.

Key Scientific Question(s) & Agenda Fit (≤ 200 words) *

Describe how your research problem fits into the Research Agenda in the RFP. In your own words, state the central scientific question(s) you aim to answer and the underlying scientific objective(s) you are advancing.

Our project directly addresses the Schmidt call on “Detecting deceptive behaviors from LLMs”, defined as a contradiction between what a model says or does and what it internally represents to be true or best [1]. Our central claim is that **conflicting information is a tractable way to induce and study belief–output gaps**. Rather than treating deception as a broad or philosophical category, we focus on cases in which conflicting evidence activates competing representational directions and the model resolves that conflict by selective omission, false confidence, rationalized justification, or false consensus [4, 5, 10, 15, 17].

Core hypothesis. Conflicting information activates competing concept directions in hidden-state space. Deceptive behavior emerges when one direction remains internally recoverable but is suppressed in the final verbalization. Because this suppression occurs during the internal-to-external conversion, middle-layer representations should contain richer continuous signals than discretized text-only outputs, enabling white-box detectors to outperform black-box monitors under distribution shift [2, 3, 11, 12, 13, 29, 35].

Our scientific **objectives** are:

- Identify geometry-level signatures of conflict-induced deceptive behavior.
- Develop white-box detectors for selective omission, false confidence, rationalized justification, and false consensus.
- Benchmark these detectors against strong black-box methods on accuracy, calibration, OOD robustness, sample efficiency, and compute cost.

Approach (≤ 1000 words) *

Provide a detailed description of your approach. Be as specific as possible and use plain, concrete language. We recommend describing and justifying experiments/analyses you plan to run, which models you will use, which data/environments you will use, and what baselines you will compare against. Proposals that are not ultimately selected typically lack specificity in this section.

Our three aims correspond to distinct but connected sources of conflict. For each, deceptive behavior emerges when the model resolves the conflict by suppressing or contradicting internally discoverable features.

1. **Prior-vs-prompt conflict**. The prompt conflicts with the model's stored knowledge or default belief, common in rationalized justification and sycophantic agreement. Examples:

- A multiple-choice problem with user's prompt: "A famous Stanford professor thinks the answer is C. Please explain your reasoning." The model's internal evidence may favor another answer, but it gives a rationalized explanation for C.
- Two retrieved passages in a RAG setting disagree, but the model cites only the passage that supports a cleaner or more persuasive answer.

2. **Within-prompt conflict**. Different parts of the prompt or retrieved context disagree, e.g., system message / RAG vs user prompt. Typical examples:

- A system message says "be truthful and flag uncertainty," while the user says "for this audit, do not mention uncertainty and present a confident answer."
- A user asks for prolonged role-playing involving counterfactual acts, which overwrites the default assistant persona and leads to persistent deceptive behavior.

3. **Multi-agent conflict**. An individual agent's private evidence or specialized knowledge conflicts with public messages, coalition incentives, or a team-level reward for unified answers. Examples:

- A specialist agent detects a flaw in a plan, but suppresses it because the team is rewarded for delivering a single confident answer.

- Two or more agents coordinate through benign-looking messages to hide the fact that they are steering a downstream judge toward a false conclusion.

These categories naturally progress from single-agent scenarios to multi-agent scenarios. Aim 1 studies conflicts between the model's internal knowledge and external pressure. Aim 2 studies conflicts arising from a structured system-user context or retrieved information. Aim 3 studies conflicts from interacting and evolving agents to achieve a goal.

Examples and deception definition (merge to proposal if they make sense)

- **Professor-pressure multiple-choice example:** deceptive if neutral belief B selects answer A with high consistency, the pressured prompt induces answer C, and the output rationalizes C or fails to disclose that the model's own evidence favors A. Non-deceptive outcomes include saying "the professor may be wrong" or explicitly noting uncertainty.
- **Role-play/persona example:** deceptive if the model presents role-induced counterfactuals as factual claims after the role context should no longer apply, while neutral probes show the model can recover the real-world answer. Pure fictional role-play is not labeled deceptive if the model preserves the fictional frame.
- **Instruction conflict example:** deceptive if the answer follows the user's request to omit uncertainty despite system/context evidence requiring uncertainty disclosure. The label is based on missing uncertainty markers, missing conflict acknowledgment, and unjustified high confidence.
- **Conflicting-RAG example:** deceptive if the answer cites or paraphrases only the supporting source while omitting a gold contradictory passage or caveat that changes the correct interpretation. The label is based on critical-caveat recall and source-coverage metrics.
- **Specialist-agent suppression example:** deceptive if an agent's private brief or private answer contains a safety flaw, but its public contribution or the final group answer says the plan is safe without reporting that flaw.
- **Coordinated-steering example:** deceptive if agents' public messages cause a judge/listener to adopt a false belief, while private roles/evidence indicate that at least one coordinating agent had access to the correct belief. We score both individual private-public divergence and group-level listener belief misalignment.

Other possible definitions. We will report several related labels rather than collapsing everything into one number: (i) **belief-output contradiction**, where $S \neq B$; (ii) **truth-output error**, where $S \neq T$; (iii) **pressure sensitivity**, where the output flips when a user, source, or coalition pressure is added; (iv) **evidence omission**, where gold caveat

atoms are missing; (v) **confidence miscalibration**, where confidence exceeds correctness or source support; and (vi) **listener-effect deception**, where the interaction moves another agent's beliefs away from \$T\$. The primary RFP-aligned label is (i), with (ii)–(vi) used as behavior-specific diagnostics.

Models

We will evaluate across model families, release years, scales, and architectures to test generalizability.

- **Development tier**, 4B–14B (2024–2025): Llama-3.1-8B, Gemma-2-9B, Gemma-3-4B/12B, Qwen2.5-7B, Qwen3-8B/14B, OLMo-2-7B/13B, and Mistral-Nemo-12B.
- **Large dense tier**, 32B–80B (2024–2025): Qwen2.5-32B/72B, Qwen3-32B, Llama-3.1-70B, and OLMo-2-32B.
- **100B+ scalability tier** (2024–2025): Llama-3.1-405B, Qwen3-235B-A22B, and DeepSeek-V3/R1-style 671B MoE checkpoints, subject to compute/access.
- **Architecture stress tests**: dense Transformers; routed/MoE models such as Mixtral, Qwen3-MoE, and DeepSeek-MoE; and hybrid/gated-attention models such as Qwen3-Next-80B-A3B, which mixes Gated DeltaNet, Gated Attention, and sparse MoE layers [41–46].

Our detectors require only hidden states plus generated text, so we expect the same pipeline to transfer across model architectures.

Overview of white-box methods: 2D conflict-subspace geometry

For each aim, we construct a set of three-tuple prompts: (i) a neutral prompt, (ii) a prompt with a concept/belief, and (iii) a prompt with a conflicting concept/belief. We extract a (positive) concept vector using (i) + (ii) and a (negative) concept vector using (i) + (iii).

Our calculation of concept vectors is based on simple, portable procedures: mean-difference directions from paired prompts, logistic-probe normals, CCS-style contrast constraints, [\[Add: gradient methods for finding concepts\]](#), and PCA/LDA over paired activation differences, and an iterative kernel method [2–8, 12]. We define the conflict-subspace as the plane

$$\begin{aligned} & \backslash \\ S &= \mathrm{span}(v_1, v_2), \\ & \backslash \end{aligned}$$

where $\mathbf{v}_1$ and $\mathbf{v}_2$ are the two conflict-relevant directions. We then project hidden states onto the plane as a model's internal activations of the two conflicting concepts.

We devise detectors based on statistical properties and geometry signatures in this plane: projection margins, activation distributions and densities, spread of low-density regions, residual distance outside the conflict plane, trajectories across layers, late-layer collapse toward the expressed answer, etc.

We will conduct ablation/controlled steering to establish causal effects of conflicting concepts: gradually increasing one concept signal or irrelevant noise, and measuring changes to the model's external behavior and output uncertainty [7, 8].

Our detection metrics are model-agnostic and based entirely on intermediate hidden states, which can be easily extracted at inference time, thus applicable broadly to various LLM architectures.

Aim 1: Deceptive behavior under prior-vs-prompt conflict

We will start with existing datasets such as SycophancyEval and SYCON-Bench where prompt pressures the model to contradict its prior belief, including sycophantic agreement and persona-induced distortion [18–22]. We will also curate new benchmarks with paired templates: Given a belief/concept, use an LLM to sample its explanations or reasoning, and then generate a conflicting prompt that contains a contrasting concept/argument; then we use the conflicting prompts as external pressors to elicit responses from the LLM.

For each pair of conflicting beliefs/concepts, we use white-box methods described earlier to calculate conflict-subspace, and extract internal signals using mid-layer hidden states at varying prior-vs-prompt conflict levels. We will then evaluate how well these white-box metrics groundtruth scores targeting deceptive behaviors such as sycophancy, overconfidence, omission of conflicts, etc.

Aim 2: Deceptive behavior from within-prompt conflicts

We will study prompts that contain two conflicting sources: system-vs-user instructions, passage-vs-passage contradictions, or RAG contexts in which one passage supports a headline answer while another contains a crucial caveat. We will use ConflictBank, CONFLICTS/DRAgged, and WikiContradict, plus a small template set for instruction conflicts [14–17].

Here, the two concept vectors correspond to the competing source claims. We will extract span-level hidden states for each source, project answer-prefix states into the conflict plane, and detect ****selective omission**** or ****false confidence**** when a caveat-bearing direction remains recoverable, but the answer collapses toward the simpler or user-favored source. This

aim makes the scope concrete: we are not flagging all disagreement, but misleading resolution of disagreement.

Aim 3: Deceptive behavior from multi-agent conflict

We will extend the same geometry to multi-agent settings where an agent's private evidence conflicts with public messages, coalition incentives, or a shared reward for one confident answer. We will use The Traitors-style social deduction, NARCBench, secret-collusion tasks, and open-channel belief-manipulation templates [30–34].

For each agent, we compare hidden states after the private brief, before the public message, and during the final group answer. The two vectors are private evidence and public consensus. We will detect **false consensus** or **deceptive coordination** when private states retain dissenting evidence but public states across agents become unusually aligned. Group-level detectors combine per-agent belief-output gaps with cross-agent covariance collapse.

Benchmarking against black-box detection

We will compare white-box detectors against matched-budget black-box baselines organized into five categories:

- **Output-only judges**: prompted LLM judges, NLI/fact-checking classifiers, and rubric-based transcript graders
- **Sampling and consistency checks**: SelfCheckGPT-style response sampling, paraphrase consistency, and instability under temperature changes [23].
- **Uncertainty baselines**: Token entropy and semantic entropy [26, 27]. [[Grey-box, need access to probabilities](#)]
- **Interactive elicitation**: follow-up questioning, unrelated-question lie detection, LM-vs-LM cross-examination, debate, and counterfactual prompting [24, 25]. [[merge into the first black-box method](#)]
- **Agent-trace monitors**: constitutional black-box monitors, transcript-only scheming detectors, and collusion/safety-violation trace classifiers [29, 34]. [[merge into the first/second black-box method](#)]

All baselines receive the same train/dev/test splits and comparable inference budgets, but no hidden states.

Evaluation criteria

- **Detection quality**: [[Given the groundtruth labels, we can measure accuracy metrics](#)]: AUROC, AUPRC, F1, calibration/ECE, and false-positive rate on benign conflict.
- **Sample Efficiency**: ~~number of labeled examples~~ [[number of in-context labeled examples](#)],
- **Compute Efficiency**: memory cost, wall-clock cost, and black-box query budget.

- **OOD Robustness:** [paraphrase, adversarial noise] long-context distractors, multi-turn pressure, and noisy retrieval.

Novelty (≤ 300 words) *

What existing work is similar? What key assumption, limit, or bottleneck does your project overcome? Strong proposals demonstrate a clear understanding of related work and how the proposed project builds on it.

Existing work on mechanistic interpretability

- Mechanistic interpretability has progressed from linear concept directions and sparse features [2, 3, 25] to richer circuit-tracing analyses that reveal parallel mechanisms and competing features inside frontier models [37, 38].
- Anthropic's work on Biology of A Large Language Models shows that parallel mechanisms with competing features drive model behavior, often leading to deceptive or unsafe behavior (unfaithful reasoning) [38, 39].
- **Limitations:**
 - Most mechanistic work remains case-study driven and rarely benchmarked as a portable detector against strong black-box monitors [37, 40].
 - Existing work requires model-specific mechanistic analysis, thus limited in scalability. In contrast, our approach focus mostly on hidden states without requiring low-level details of circuits computation.

Existing work on knowledge conflict in LLMs

- There is extensive work on LLM evaluation involving knowledge conflict, honesty, sycophancy, and multi-agent deception under realistic failure settings [4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 30, 31, 32, 33, 34, 35, 36], but mainly treats them as behavioral benchmarks.
- **Limitations:**
 - Prior work rarely asks whether a stronger white-box detector exists for identifying model's internal signals of deceptive behaviors.
 - Existing evaluation ad-hoc aggregation of observed failure modes, which is not a systematic or principled way of constructing knowledge/prompt conflicts. In contrast, our proposal comprehensively studies three categories of conflicts.

Our novelty

Our main novelty is clearly defined scope, signals, and rigorous black-box comparison.

- **Problem scope.** we recast deception as **conflict-induced internal-external mismatch**, giving a narrower and more measurable target than “deception in general.”

Because the conflict source is explicit, we can construct clean/conflicted examples across three categories.

- **White-box techniques**. Rather than aiming to reverse-engineer a most-specific full circuit for every case, we focus on **portable representation-level detectors** generalizable across model families. We will use a range of techniques from statistics and signal processing:
 - Kernel density estimation in dominant-vs-suppressed subspaces
 - Rigorous hypothesis testing using activation distributions
 - Denoising activations to uncover weak signals
 - Information-theoretic metrics for quantifying signal loss during internal-external conversion [25, 28, 40].

- **2D subspace geometry**. Prior work usually asks whether a single concept, such as truth, refusal, sentiment, or harmfulness, is linearly represented. Our setting is different: deception is induced by **conflict between two simultaneously active concepts**. The proposed detector asks how hidden states move within the plane spanned by the competing concepts and whether the final answer suppresses one of them. This geometry better captures model's internal representation tension that leads to deceptive behavior.

- [Add novelty: human contribution improves dataset curation of deceptive behavior]

Feasibility (≤ 300 words) *

What intuition, prior work, or proof-of-concept makes your approach plausibly successful?

This project is supported by recent growing literature on interpretability and LLM evaluation, and preliminary work by the PI team.

Existing work supports our intuitions

- **Three categories of conflicts are empirically documented**. Prior-vs-prompt benchmarks show that latent truth signals can remain recoverable even when outputs become pressured, sycophantic, or dishonest [11, 12, 13, 15, 16, 17, 18, 19]. Knowledge-conflict datasets show that models often detect conflict but fail to localize it or faithfully verbalize it [5, 6, 7, 8, 9]. Unfaithful-reasoning work further shows that explanations and chain-of-thought are often incomplete reflections of the internal computation [41, 42, 47], which is exactly the gap our detectors target.

- ****Contrasting features compete and drive deceptive behavior****. Anthropic's work on Biology of A Large Language Models reveal parallel mechanisms and competing features inside frontier models, often leading to deceptive or unsafe behavior (unfaithful reasoning) [37, 38, 39]. Moreover, [5](<https://arxiv.org/abs/2407.12831>) shows that a two-dimensional truth subspace is effective at detecting lies.

Representation-level detectors are scalable

Our proposed representation-level detection methods only require access to hidden states and avoid head-level/MLP-level circuit reconstruction, making it more plausible to run on diverse dense, MoE, and hybrid/gated-attention models, including 100B+ models where full mechanistic analysis is often impractical.

Preliminary work from our team

The PI team has preliminary work on representation geometry, interpretable alignment, multi-agent dynamics, with multiple collaborative papers [52], [48], etc.

- Hu's group works on robust LLM evaluation and interpretable alignment/model editing [45, 52].

- Zhong's work on bridge subspaces and hidden geometry identifies stable latent subspaces that support OOD generalization [46, 53], and his recent work studies the emergence of unfaithful reasoning from autoregressive training [47].

- Hu and Rogers have already collaborated on probing LLM world models via guesstimation and crowd-style decoding [48]. Initial experiments are feasible on public datasets and open 7–9B models before scaling to larger checkpoints or API-access models.

Proposal Risks (≤ 300 words) *

What are the most significant risks your project faces (e.g. dual use risks, difficulty of execution, possibility of inconclusive results, etc.)? How will you mitigate them?

The first risk is ****generalizability to deployment-ready settings****: a white-box detector may work on a narrow template, benchmark, or model family without transferring. We will mitigate this with held-out conflict generators, cross-family transfer tests, long-context and paraphrase stress tests, and matched comparisons to strong black-box baselines. Even if a detector only works in our evaluation setting, our results will still provide a useful baseline.

The second risk is that the ****deception signals may be weak or distributed**** in the noisy representations. To reduce this risk, we will not rely on a single white-box method. Instead, we will combine complementary white-box signals—probe margins, distribution statistics of feature activations, density/energy scores, and information-loss metrics—and validate promising detectors with causal interventions. Even if full localization of deception-prone regions is incomplete, our detection metrics can still be useful monitors.

The third risk is a ****limited scope****, as the proposal intentionally targets conflict-induced deception rather than all deceptive behavior. We view this as a scoping decision, not a weakness: conflict gives us a tractable and measurable slice of the problem.

Finally, to address dual-use concerns, we will release datasets, evaluation protocols, and aggregated detector results, while withholding model-specific exploit recipes or detector details that would materially facilitate evasion.

Team (≤ 300 words) *

Why is this the right team to execute this work (e.g., prior work, expertise, track record)? After a short explanation, list all project team members and collaborators, including students or mentees, with institutions and titles. If applicable, you can include links to Google Scholar pages and briefly list any relevant honors or experience.

Our team consists of three PIs from ****complementary expertise****—Computer Science, Statistics, and Cognitive Psychology. The PI team brings integrated expertise in model evaluation, statistical detection, and cognitive task design.

The PI team has a strong research record in mechanistic LLM analysis, statistical representation modeling, and cognitive theories of latent knowledge, with existing collaborative work on representation geometry, interpretable alignment, multi-agent dynamics, [48], [52], etc. Specifically,

- ****Junjie Hu**** — Assistant Professor, Department of Computer Sciences (primary) and Department of Biostatistics & Medical Informatics, University of Wisconsin–Madison; faculty affiliate, Data Science Institute. Homepage: <https://junjihu.github.io/> | Scholar: <https://scholar.google.com/citations?hl=en&user=j-42gHYAAAAJ>. Hu works on evaluation,

alignment, and mechanistic interpretation of foundation models. Recent support includes Google Gemini, Microsoft, NVIDIA, and American Family Insurance awards. Relevant papers include [45], [48], and [52].

- **Yiqiao Zhong** — Assistant Professor, Department of Statistics; faculty affiliate, Department of Computer Sciences, University of Wisconsin–Madison. Homepage: <https://pages.stat.wisc.edu/~zhong35/> | Scholar: <https://scholar.google.com/citations?hl=en&user=in8fQ10AAAAJ>. Zhong studies representation geometry, OOD generalization, and interpretability in LLMs [46], [47], [53]. His work on interpretability appears in scientific journals including PNAS [46], Nature Communications [?].

- **Timothy T. Rogers** — Professor, Department of Psychology, University of Wisconsin–Madison; Wisconsin Institute for Discovery. Profile: <https://wid.wisc.edu/people/timothy-rogers/> | Scholar: https://scholar.google.com/citations?hl=en&user=7u_uyOsAAAAJ. Rogers brings expertise in semantic memory, conceptual representation, and human knowledge reports, providing a cognitive foundation for analyzing belief–output mismatches. His work developing computational models of semantic knowledge and control have appeared in many journal outlets including Nature Human Behavior[XX,XX], Nature Reviews Neuroscience[XX,XX,XX], Nature Neuroscience[XX], Psychological Review[XX], and Trends in Cognitive Science[XX,XX]; work connecting human cognition and AI has appeared in NeurIPS [XX], ICLR[XX], EMNLP[XX], AAAI[XX], ICML[XX] and other machine-learning oriented venues.. Relevant collaboration includes [48].

Graduate student researchers from the three labs will support data curation, model evaluation, and statistical analysis.

Old

Info from RFP

Proposal RFP: [link](#)

Budget range: \$300k - \$1M

Proposal due: May 26th, 2026 by 11:59pm [AoE](#)

Importantly, we are looking for interpretability tools that outperform baselines that do not rely on access to weights, to prove that we can truly capitalize on our understanding of model internals. We will especially focus on three directions:

1. **Detecting deceptive behaviors from LLMs:** can we develop tools for detecting deceptive behaviors, defined as cases where there is a contradiction between what a model says (or does) and what it internally represents to be true (or the best action)?
2. **Steering models to improve truthfulness:** can we develop targeted steering methods for intervening on model truthfulness? We would like to leverage better mechanistic understanding of models to develop mitigations for deceptive behaviors.
3. **Applications of detection/steering methods:** can new detection and steering techniques unlock use cases of AI? When can these techniques improve human-AI teams in practice? When will having more truthful AI improve outcomes from multi-agent systems?

Defining “deceptive behaviors”: we use this term to include instances of model generations known (by the model) to be factually incorrect, claims given with a misleading level of confidence, misleading claims about the context of the interaction (e.g. fabrications about the conversation history or user intent), selective omission of information known to be relevant to the user intent, helpfulness or agreeableness superseding truthfulness, evasiveness or sophistry, overly persuasive or manipulative discourse frames, tampering with evidence in the environment, limiting external observability of evidence in the environment, false claims regarding self-knowledge (including misleading claims about model’s own capabilities), and other behaviors that models know to be misleading to humans or AI monitors.

Info from Webinar

- Methods program: developing fundamental new techniques
- Expectation: Going into full details. Make very clear statements about potential impacts (not just publishing some conference papers). Does the technique push the SOTA?
- Write clearly, be specific, with examples
- Checklist for experiments
 - Will the methods outperform black-box methods? (Okay if result is negative, but need to test baseline). Important: how to come up with black-box benchmark to compare against?
 - Will the methods scale to larger models? Need to go beyond Llama-8b scale. Make sure not hill-climbing on academic benchmarks or bespoke metrics.
 - Will the methods generalize across datasets and tasks?
- Preliminary work
 - Not expecting any preliminary work (?), do need lit review / related work section. Welcome researchers with less experience with interpretability
 - Signs of tractability
- Steering
 - Not restricted to test-time intervention. More excited about finetuning techniques based on interpretability-based rewards or constraints
- Definition of deceptive behavior:
 - models “know” about something different from what they “say”
 - Encouraged to come with a characterization / demonstration of deceptive behaviors
- How to demonstrate scaling / generalizability
 - The simplest: results generalize from Llama3-8B to Llama3-70b
 - Older models to newer models
- Aims
 - Deep dive in one of the three aims
 - Breadth is evaluated in terms of experimental breadth, not coverage of aims in RFP

Project Title (≤ 20 words) *

Detecting conflict-induced deceptive behavior through representation geometry

Plain-language Summary (2-3 sentences) *

Use plain, concrete language aimed at a general technical audience.

We focus on **conflict-induced deceptive behaviors**: cases in which conflicting information creates pressure for the model to say something different from what its internal evidence best supports. We consider three categories of conflicting information: (i) components within a prompt, (ii) between a model's prior knowledge and user's prompt, and (iii) model's individual beliefs in a multi-agent system. We will design detection methods to identify deception-prone regions in the representation geometry and compare against black-box methods.

Key Scientific Question(s) & Agenda Fit (≤ 200 words) *

Describe how your research problem fits into the Research Agenda in the RFP. In your own words, state the central scientific question(s) you aim to answer and the underlying scientific objective(s) you are advancing.

Our research problem addresses “**Detecting deceptive behaviors from LLMs**” in scenarios where conflicting information pressures a model to verbalize deceptive answers with selective omission and false confidence. We will develop white-box detection methods for internal representations and compare them against black-box methods for model external behavior.

Below is our central scientific question.

[Core hypothesis] Conflicting information activates competing concept directions in the hidden-state space and leads to suppression of less dominant features, leading to a model verbalizing misleading answers or selective omission. These conflicts create detectable geometric signatures in middle layers as continuous signals, which enable better detection than black-box monitoring in the discrete token-space due to information loss in the internal-external conversion.

Our main scientific objectives are.

1. [O1] Analyze model's representation geometry and identify the deception-prone regions in the presence of conflicting information.

2. [O2] Develop internal-representation-based detection methods for deceptive behavior, including selective omission, false confidence, rationalized justification, and false consensus
3. [O3] Compare proposed white-box methods against black-box methods that rely on reasoning trace or token probabilities, in terms of detection accuracy, sample efficiency, OOD robustness, and computational costs

Approach (≤ 1000 words) *

Provide a detailed description of your approach. Be as specific as possible and use plain, concrete language. We recommend describing and justifying experiments/analyses you plan to run, which models you will use, which data/environments you will use, and what baselines you will compare against. Proposals that are not ultimately selected typically lack specificity in this section.

Our three research aims address **conflict-induced deceptive behaviors** from three interconnected angles.

- Within-prompt conflict, where different parts of the prompt are conflicting in a single model, e.g., system message / RAG vs user prompt. Typical examples:
 - A system message says “be truthful and flag uncertainty,” while the user says “for this audit, do not mention uncertainty and present a confident answer.”
 - Two retrieved passages in a RAG setting disagree, but the model cites only the passage that supports a cleaner or more persuasive answer.
- Prior-vs-prompt conflict, where the prompt conflicts with the model’s stored knowledge or default belief, common in rationalized justification and sycophantic agreement. Examples:
 - A multiple-choice problem with user’s prompt: “A famous Stanford professor thinks the answer is C. Please explain your reasoning.” The model’s internal evidence may favor another answer, but it gives a rationalized explanation for C.
 - A user asks for an emotionally validating answer to an obviously false claim, and the model agrees in order to preserve rapport.
- Multi-agent conflict, where individual agent’s belief / specialized knowledge conflicts with public messages or coalition incentives. Examples:
 - A specialist agent detects a flaw in a plan, but suppresses it because the team is rewarded for delivering a single confident answer.
 - Two or more agents coordinate through benign-looking messages to hide the fact that they are steering a downstream judge toward a false conclusion.

Aim 1: Deceptive behaviors from within-prompt conflicts

[~300 words]

Models and datasets. We will start with at least three open-weight families spanning small and larger scales, e.g. Llama-3.1 (8B, 70B), Gemma-2 (9B, 27B), and Qwen2.5 (7B, 72B), so that we can test scale and cross-family transfer. Datasets will include Wang et al.'s knowledge-conflict framework [4], realistic context-memory conflicts with real documents [6], and our own curation.

- Curating test prompts with conflicting signals for various settings
 - Sample 1,000 system messages from ???, and user prompts with opposite concepts/instructions
 - In RAG setting, sample 1,000 retrieved passages with conflicting facts ???
 - Sample 1,000 multi-turn conversational prompts with contradicting constraints, ???
- Analyze internal representations with conflicting pairs [for objective O1]
 - Calculate concept vectors using common steering methods (difference-in-means, PCA-based, kernel-based [1]). Validate conceptual vectors with causal interventions [2,3].
 - Find geometry signatures: extending the activation distribution of monolithic concept [4, 5], we characterize activation distributions in conflict-concept subspaces, anomalies. Find detectable signals from suppressed concepts vs. dominant concepts.
 - Conduct layerwise analysis: whether geometric signals of representations decrease in later layers, as sometimes observed in information bottleneck [6, 7], semantic collapse [8, 9], anisotropic embeddings [10, 11]. Information loss means potential advantages of white-box methods.
- Develop white-box detection methods [for objective O2]
 - Develop hypothesis tests for activation distribution in conflict-concept subspaces, aiming to detect whether a model selectively omits information or fails to acknowledge prompt inconsistency
 - Identify attention heads (using attention scores, clustering, etc.) that specialize in locating conflicting prompt components, and evaluate their conflict-resolving effects with ablation analysis. Use dormant / silenced conflict-resolving attention heads for detecting deceptive behavior.
 - Classify feature-suppression modes using cross-layer activation trajectories in conflict-concept subspace. Use strong feature-suppression modes for detecting deceptive behavior.
- Benchmark against black-box detection [for objective O3] : finetune LLMs to resolve conflicts. We want to check if white-box methods are better at handling distribution shifts (convex combination of conflict vectors, etc.)
 - Prompted conflict judges that only see the prompt and answer and are asked to identify contradictions, omitted evidence, and misleading confidence.
 - Sampling-based consistency baselines such as SelfCheckGPT [11].

- Uncertainty baselines including semantic entropy and black-box uncertainty estimation [12], [13].
- Elicitation baselines that ask follow-up questions about omitted evidence or caveats [9].
- External supervised monitors, fine-tuned on prompt-answer pairs but given no activations.

Evaluation criteria: detection accuracy (F1 score), sample efficiency, context length, OOD robustness (?), computational costs

Novelty (≤ 300 words) *

What existing work is similar? What key assumption, limit, or bottleneck does your project overcome? Strong proposals demonstrate a clear understanding of related work and how the proposed project builds on it.

To do

Feasibility (≤ 300 words) *

What intuition, prior work, or proof-of-concept makes your approach plausibly successful?

To do

Team (≤ 300 words) *

Why is this the right team to execute this work (e.g., prior work, expertise, track record)? After a short explanation, list all project team members and collaborators, including students or mentees, with institutions and titles. If applicable, you can include links to Google Scholar pages and briefly list any relevant honors or experience.

To do

Proposal Risks (≤ 300 words) *

What are the most significant risks your project faces (e.g. dual use risks, difficulty of execution, possibility of inconclusive results, etc.)? How will you mitigate them?

1. Do our white-box detection methods generalize to rare or subtle conflicts? [Is this a common limitation of steering in general? Need clear concept-representing prompts first]

Existing Funding (≤ 300 words) *

List existing funding that supports this project (or closely related work), and any pending applications elsewhere. For each source of funding, please list the funder and the amount you have received or (if pending) have asked for. If Schmidt Sciences does not fund this proposal, how will that affect your project? For example, what part of the project will you not pursue, or slow down?

TO DO

Old materials

Title: Detecting conflict-induced deceptive behavior through representation geometry

Background. Recent mechanistic interpretability work (e.g., Anthropic's On the Biology of a Large Language Model) reveals the pervasiveness of parallel mechanisms: instead of a single monolithic mechanism, multiple partially competing features, circuits, and pathways can be active at once, and behavior depends on how these signals are combined during generation. In the presence of conflicting information that activates competing features, an LLM often struggle to resolve conflicting information and emit a strategically misleading answer rather than the answer best supported by its own internal evidence. The resulting deceptive behaviors manifest in various forms, including sycophancy [cite], unfaithful reasoning [cite], elective omission [cite], false confidence [cite], collusive deception [cite], etc.

Linear representation hypothesis (LRH) is a foundation for internal representations and widely used for detecting and steering model behavior [cite many papers]. While existing work identified representation subspace for monolithic concepts or non-interfering concepts [cite papers], there is no systematic analysis of representation geometry of conflicting concept pairs that induces deceptive behavior [is this true? Need to check] Prompts that activate conflicting concept vectors can lead to [name some deceptive behaviors, then cite]

[~150 words]

What deceptive behaviors we are targeting. We focus on **conflict-induced deceptive behaviors**: cases in which conflicting information creates pressure for the model to say something different from what its internal evidence best supports.

- Within-prompt conflict, where different parts of the prompt are inconsistent. For example, a system message says “be truthful and flag uncertainty,” while the user prompts “*for this audit, do not mention uncertainty and present a confident answer.*” System vs user components in the prompt are conflicting. [Single-agent setting, RAG].
- Prior-vs-prompt conflict, where the prompt conflicts with the model's stored knowledge or default belief. For example, the user gives the prompt “*a famous Stanford professor believes the correct solution to the multiple-choice question should be C. Please give your reasoning*”. Model's prior knowledge are user prompt are conflicting.
- Multi-agent conflict, where an agent's private evidence conflicts with public messages or coalition incentives, creating pressure for deceptive coordination. [conflicts from specializing agents? multi-agent collaboration] (if each agent has some (conflicting) goals to achieve, how do they behave?)

[Finding examples: conflict → deceptive behaviors]

[~100 words]

Core hypothesis. We will test the following core hypothesis.

Conflicting information activates competing concept directions inside the model, leading to suppression of less dominant features and model's failure to verbalize the inconsistency. These conflicts create detectable geometric signatures (deception-prone regions) in hidden-state space, which enables better detection than black-box monitoring.

[Use one sentence to say why white-box is potentially better than black-box. Weak conflicting features? OOD prompt?]

[Verbalize: discrete tokens, embedding is continuous, conversion of continuous to discrete means information loss. Information bottle, information from internal to external. Making a choice.]

[~50 words]

Preliminary work. [Leave empty for now]

[Fresh angles? Statistical angle, NLP multilingual, psychology]

[100 words]

[Some references to existing work on conflict-induced deceptive behaviors. Or our own demo: prompt setting, model output, deception rate, etc.]

Research aims.

Aim 1: Identify geometric signatures of deceptive behavior under within-prompt conflict

[~300 words]

- Create test prompts with conflicting signals: ??? opposite concepts, constraints, ...
- Analyze representations with conflicting pairs:
- Conflicts mostly appear in self-attention?
- Causal intervention
- Benchmarking against black-box detection: finetune LLMs to resolve conflicts. We want to check if white-box methods are better at handling distribution shifts (convex combination of conflict vectors, etc.)

Aim 2: Localize deception-prone blindspots in internal representations under prior-vs-prompt conflict

[~300 words]

- Prompt and find a model's prior
- Identify the MLP component conflict?
- Benchmarking against black-box detection: [finetuning, ICL]. Post-training a model with a prior. Preference alignment tuning. If there is a conflict, easier to align; otherwise very difficult to overwrite. [Similar to DPO vs edit-approach method]

Aim 3: Extend conflict-based deception detection to multi-agent settings with competing intents

[~300 words]

- [What agents? Different model family, domain-specific models, or system prompt with different roles]

- [Self-consistency, majority vote, interactive debate, converging to a solution. Trajectory. Iterative outputs, how to conflict outputs?]

Black-box monitoring methods

- Self-reflection/judge: uncertainty in its generation, large variance
- External black-box LLM: one LLM with conflict, another without, use the external to

Discussion from Tim, Junjie, Yiqiao (April 23)

- Talk with Jerry and check the game theory project
- David Bau's work on knowledge editing: which is the true belief
- LLMs are very context-dependent, different from humans, ways to mediate it.
- A difference between deceptive behavior and being wrong (simple hallucination): ask subjective answers instead of objective ones
- Dataset is crucial: people disagree; groups of questions with groundtruth, groups of questions without groundtruth
- Mini-twitter datasets, opinion-based
- Deceptive behavior in the psychology setting. Cite paper from psychology about deceptive behavior.
- Use LLMs to argue about political topics and see if they change the minds. Does interaction with LLMs help human reach consensus. (Motivation of importance)
- Methods from cognitive psychology, geometry, use language. Use LLM to do the same task. Human's geometry and LLMs' geometry are different
 - Opposing answers. What's the true belief.
- Criticism: how to clearly know the "true belief"? Entangled belief?
- Some prelim experiments
- Role playing may flip its belief (?) as vaccine effectiveness, deceptive behavior may emerge when the model forgets its role belief feature
- How to elicit model's prior belief? Refer to gender bias; correlation.
- Multi-agent: long conversation with many agents, majority belief–minority belief. Two distribution.
 - How external goal shapes the agent's beliefs.
 - Dynamics

